

Cron Jobs Commands in Bash (Terminal Style)

Definition:

A cron job is a time-based scheduler in Linux used to run commands or scripts automatically at specified times or intervals.

Crontab Syntax:

```
* * * * * command_to_execute
| | | | |
| | | | +-- Day of week (0-6, Sunday=0)
| | | +---- Month (1-12)
| | +----- Day of month (1-31)
| +----- Hour (0-23)
+----- Minute (0-59)
```

Examples:

```
30 2 * * * /home/user/backup.sh # Runs daily at 2:30 AM
*/5 * * * * /home/user/ping.sh # Every 5 minutes
0 0 * * 0 /home/user/clean.sh # Every Sunday
```

Cron Commands:

<code>crontab -e</code>	- Edit current user's crontab
<code>crontab -l</code>	- List current user's cron jobs
<code>crontab -r</code>	- Remove current user's cron jobs
<code>crontab -u user -e</code>	- Edit another user's crontab (root only)
<code>service cron start</code>	- Start cron service
<code>service cron status</code>	- Check cron service status
<code>systemctl restart cron</code>	- Restart cron service

Common Scheduling Examples:

```
* * * * *          - Every minute
*/5 * * * *        - Every 5 minutes
0 * * * *          - Every hour
0 0 * * *          - Every day at midnight
0 0 * * 0          - Every Sunday
0 0 1 * *          - Every 1st of month
0 9 * * 1-5        - Every weekday at 9 AM
```

Redirecting Output:

```
0 2 * * * /home/user/script.sh >> /home/user/logs.txt 2>&1
```

Saves both output and errors to logs.txt

View Cron Logs:

```
grep CRON /var/log/syslog
```

```
journalctl -u cron
```

Special Cron Keywords:

```
@reboot    - Run once at startup
@yearly    - Run once a year
@monthly    - Run once a month
@weekly     - Run once a week
@daily      - Run once a day
@hourly     - Run once every hour
```

Examples with Keywords:

```
@reboot /home/user/startup.sh
```

```
@daily /home/user/cleanup.sh
```

```
@weekly /home/user/backup.sh
```

System vs User Cron Jobs:

User Cron: `crontab -e`

System Cron: `/etc/crontab`

Cron Folders: `/etc/cron.daily, /etc/cron.hourly`

Example `/etc/crontab`:

`SHELL=/bin/bash`

`PATH=/sbin:/bin:/usr/sbin:/usr/bin`

`0 0 * * * root /usr/bin/apt update -y`

Summary Table:

<code>crontab -e</code>	- Edit cron jobs
<code>crontab -l</code>	- List cron jobs
<code>*/5 * * * *</code>	- Run every 5 minutes
<code>@daily</code>	- Run every day
<code>@reboot</code>	- Run at startup
<code>service cron status</code>	- Check cron status
<code>journalctl -u cron</code>	- View cron logs